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Partitioning changes in ecosystem productivity by effects of species interactions in biodiversity experiments

Jing Tao, Charles A. Nock, Eric B. Searle, Shongming Huang, Rongzhou Man, Hua Yang, Grégoire T. Freschet, Cyrille Violle, Ji Zheng

Jilin Provincial Academy of Forestry Sciences, Changchun, China • Department of Renewable Resources, Faculty of Agriculture, Life, and Environmental Sciences, University of Alberta, Edmonton, Alberta, Canada, T6G 2H1 • Ontario Ministry of Natural Resources and Forestry, Ontario Forest Research Institute, 1235 Queen Street East, Sault Ste. Marie, ON, P6A 2E5 Canada , , • Forestry Division, Department of Agriculture and Forestry, Government of Alberta, Edmonton, Alberta, Canada T6H 5T6 • College of Forestry, Beijing Forestry University, China • Station d'Ecologie Théorique et Expérimentale, CNRS, 2 route du CNRS, 09200 Moulis, France • CEFE, Univ Montpellier, CNRS, EPHE, IRD, Montpellier, France • School of Agriculture and Biology, and Shanghai Urban Forest Ecosystem Research Station of National Forestry and Grassland Administration, Shanghai Jiao Tong University, China

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Abstract

Species interactions affect ecosystem productivity. Positive interactions (resource partitioning and facilitation) increase productivity while negative interactions (species interference) decrease productivity relative to the null expectations defined by monoculture yields. Effects of competitive interactions (resource competition) can be either positive or negative. Distinguishing effects of species interactions is therefore difficult, if not impossible, with current biodiversity experiments involving mixtures and full density monocultures.

To partition changes in ecosystem productivity by effects of species interactions, we modify null expectations with competitive growth responses, i.e., proportional changes in individual size (biomass or volume) expected in mixture based on species differences in growth and competitive ability. We use partial density (species density in mixture) monocultures and the competitive exclusion principle to determine maximum competitive growth responses and full density monoculture yields to measure species ability to achieve maximum competitive growth responses in mixture. Deviations of observed yields from competitive expectations represent the effects of positive/negative species interactions, while the differences between competitive and null expectations reflect the effects of competitive interactions.

We demonstrate the effectiveness of our competitive partitioning model in distinguishing effects of species interactions using both simulated and experimental species mixtures. Our competitive partitioning model enables meaningful assessments of species interactions at both species and community levels and helps disentangle underlying mechanisms of species interactions responsible for changes in ecosystem productivity and identify species mixtures that maximize positive effects.

eLife assessment

This work proposes that positive biodiversity-ecosystem functioning relationships found in experiments have been exaggerated because commonly used statistical analyses are flawed. As an alternative, the authors suggest a new analysis based on species competitive responses. Unfortunately, the presented methods are not reproducibly described, not yet complete, and **inadequate** for hypothesis testing. The reviewers agreed that the authors have either misinterpreted or chosen not to take into account much of the current research literature in the field of plant competition and biodiversity research.

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Species interactions and ecosystem productivity

Detecting changes in ecosystem productivity with biodiversity and understanding how these changes are affected by mechanisms of species interactions has been a central focus in ecology for decades (Loreau, 2000; Loreau and Hector, 2019; Mahaut et al., 2020). However, effective methods are lacking (Drake, 2003; Forrester and Pretzsch, 2015; Jaillard et al., 2018; Barry et al., 2019; Loreau and Hector, 2019; Bourrat et al., 2023). The method of additive partitioning was developed from the covariance equation to quantify changes in ecosystem productivity by comparing species performance in mixture with those expected from their monocultures using a replacement design (de Wit, 1960; Loreau and Hector, 2001; Barry et al., 2019). The resultant changes in ecosystem productivity (commonly referred to as net biodiversity effects) are mathematically decomposed into additive components, namely the complementarity effect (CE) measuring community average response and the selection effect (SE) measuring interspecies differences (Loreau and Hector, 2001; Forrester and Pretzsch, 2015; Loreau and Hector, 2019). CE ad SE are widely used to examine mechanisms of species interactions responsible for biodiversity effects, i.e., CE for positive interactions and SE for competitive interactions (Loreau and Hector, 2001; Petchey, 2003; Polley et al., 2003; Montès et al., 2008; Barry et al., 2019; Hagan et al., 2021; Feng et al., 2022), despite that neither of the additive components can be linked to specific effects of species interactions (Petchey, 2003; Hooper et al., 2005; Montès et al., 2008; Loreau et al., 2012; Forrester and Pretzsch, 2015; Loreau and Hector, 2019; Hagan et al., 2021; Bourrat et al., 2023).

Among the interspecific interactions in plants, positive interactions (resource partitioning and facilitation or collectively referred to as complementarity) increase productivity and negative interactions (species interference or interference competition) decrease productivity relative to the null expectations, i.e., species yields expected in mixture based on composition (relative yield) and full density monoculture yields (Loreau, 2000; Loreau and et al., 2005; Barry et al., 2019). Uncertainty comes with effects of competitive interactions (i.e., resource competition) that are positive for more competitive species and negative for less competitive species (Aschehoug et al., 2016; Barry et al., 2019). For example, when a highly competitive and less competitive species are mixed, more resources flow to the more competitive species. The yield gain of more competitive species from increases in resource availability (resources per capita) often exceeds the yield loss of less competitive species from decreases in resource availability (Huston, 1997; Tilman et al., 1997; Wardle, 1999; Montazeaud et al., 2018; Pillai and Gouhier, 2019), resulting in positive deviation from the total expected yield of both species (i.e., positive biodiversity effect). Competitive interactions can also result in negative biodiversity effects if the less competitive species suffers more than the more competitive species benefits.

The positive effects of competitive dominance on ecosystem productivity can be illustrated with species mixture data. In the situations of both simulated and experimental mixtures with two species, the partial density monoculture yield of the more competitive species exceeded the total expected yield of both species in 32 of the 35 mixtures (***Table 1***), meaning that a higher productivity expected for a given species mixture under the null hypothesis can be generally achieved by removing less competitive species from the mixture (***Table S1; Table S2***). In other words, the positive biodiversity effects of mixtures detected through the null expectation can be obtained from the monocultures of more competitive species at partial (lower) densities. These positive biodiversity effects, however, result from competition-induced changes in resource availability, i.e., the yield gain of more competitive species from all resources in the mixture exceeds the total yield loss expected for the less competitive species, and are not related to effects of positive species interactions. Competitive interactions are the predominant type of interspecific relationships in plants and occur in all mixtures where constituent species are competitively different (Goldberg and Barton, 1992; Pillai and Gouhier, 2019). Reporting positive biodiversity effects, without indication of possible contributions from competitive interactions, can lead to unrealistic expectations for effects of positive interactions.

Given the limits of the additive partitioning in deciphering mechanisms of changes in ecosystem productivity (Loreau and Hector, 2001; Petchey, 2003; Forrester and Pretzsch, 2015), some alternative methods have been explored to identify underlying mechanisms. This includes the use of metrics such as relative yield totals (Hector, 1998; Roscher and Schumacher, 2016) and transgressive overyielding or superyielding (Drake, 2003; Forrester and Pretzsch, 2015; Špaèková and Lepš, 2001), the multiplicative partitioning or statistical analyses to separate effects of species interactions and species composition (Kirwan et al., 2009; Jaillard et al., 2018, 2021; Chen et al., 2020), the identification of relative species influences via measurement of competitive interactions (Montès et al., 2008; Mahaut et al., 2020; Brun et al., 2022), or influences of species diversity, composition, and density on net biodiversity effects (Polley et al., 2003; Roscher and Schumacher, 2016; Jaillard et al., 2018; Mahaut et al., 2020; Tatsumi and Loreau, 2023). However, although these methods may indicate the presence of strong positive or competitive interactions or connections of the additive components with niche and fitness differences (Carroll et al., 2011; Turnbull et al., 2013; Godoy et al., 2020), they are not capable of quantifying changes in ecosystem productivity by species interactions at species or community level.

Here, we present the competitive partitioning model, a new framework for assessing effects of species interactions on ecosystem productivity based on null expectations and competitive growth responses, the proportional changes of individual size (biomass or volume) from full density monocultures to mixtures expected from species differences in growth and competitive ability (***Figure 1***). Current biodiversity experiments are generally established based on a replacement design with equal density across mixtures and monocultures (Sackville Hamilton, 1994) and do not provide data for estimating competitive growth responses. Thus, we applied this new methodological approach to simulated data generated with the GYPSY model (Huang et al., 2009) for trembling aspen (*Populus tremuloides* Michx.) and white spruce (*Picea glauca* (Moench) Voss) tree mixtures and greenhouse experimental data from grassland mixtures (Mahaut et al., 2020) (***Table 1***). We used partial density (species density in mixture) monocultures and the competitive exclusion principle to determine maximum competitive growth responses and full density monoculture yields to measure species ability to compete for resources and achieve maximum competitive growth responses in mixture. Our objectives are to (1) demonstrate the power of this new framework in distinguishing effects of species interactions responsible for changes in ecosystem productivity, (2) illustrate how the mechanisms of changes in ecosystem productivity differ between the assessments with competitive and additive partitioning models, and (3) assess the effects of competitive interactions on ecosystem productivity relative to those of other species interactions.

Table 1.

Simulated and experimental species mixtures. BPM (biodiversity effects of partial density monocultures) are calculated from the differences between the partial density monoculture yield of more competitive species and total yield expected from species relative yields and full density monoculture yields (see competitive exclusion in ***Table S1*** and ***Table S2*** for detailed calculations).

Species mixture (species1:species2)	Age	Total density	Mixture composition		Full density monoculture yields		Observed yields in mixtures		Partial density monoculture yields		
			Species1	Species2	Species1	Species2	Species1	Species2	Species1	Species2	BPM
Simulated mixed trembling aspen and white spruce in western Canada generated with GYPSY model (Huang et al., 2009) (density and stand volume yield, stems and m^3 per hectare)											
Populus:Picea	20 years	11,000	0.9	0.1	23.7	0.7	22.3	0.1	23.3	0.5	1.9
Populus:Picea	20 years	11,000	0.7	0.3	23.7	0.7	19.0	0.2	22.5	0.6	5.7
Populus:Picea	20 years	11,000	0.5	0.5	23.7	0.7	15.1	0.3	21.4	0.6	9.2
Populus:Picea	20 years	11,000	0.3	0.7	23.7	0.7	10.6	0.4	19.8	0.7	12.2
Populus:Picea	20 years	11,000	0.1	0.9	23.7	0.7	5.0	0.5	16.9	0.7	13.9
Populus:Picea	40 years	3,600	0.9	0.1	110.4	27.2	104.5	3.4	108.3	15.5	6.3
Populus:Picea	40 years	3,600	0.7	0.3	110.4	27.2	90.3	7.6	103.5	20.7	18.1
Populus:Picea	40 years	3,600	0.5	0.5	110.4	27.2	72.4	12.1	97.0	23.4	28.2
Populus:Picea	40 years	3,600	0.3	0.7	110.4	27.2	49.6	17.5	87.4	25.3	35.2
Populus:Picea	40 years	3,600	0.1	0.9	110.4	27.2	20.6	23.9	70.6	26.7	35.1
Populus:Picea	60 years	1,800	0.9	0.1	189.7	88.3	180.5	9.6	186.1	47.0	6.6
Populus:Picea	60 years	1,800	0.7	0.3	189.7	88.3	158.1	22.4	177.3	64.3	18.0
Populus:Picea	60 years	1,800	0.5	0.5	189.7	88.3	128.1	38.1	164.8	73.7	25.8
Populus:Picea	60 years	1,800	0.3	0.7	189.7	88.3	85.4	58.5	144.8	80.5	26.1
Populus:Picea	60 years	1,800	0.1	0.9	189.7	88.3	28.6	83.0	109.2	85.9	10.8
Populus:Picea	80 years	1,100	0.9	0.1	240.2	143.1	229.6	13.7	235.6	73.3	5.1
Populus:Picea	80 years	1,100	0.7	0.3	240.2	143.1	203.7	33.0	224.1	102.0	13.1
Populus:Picea	80 years	1,100	0.5	0.5	240.2	143.1	167.3	58.2	207.6	117.8	15.9
Populus:Picea	80 years	1,100	0.3	0.7	240.2	143.1	110.3	94.0	179.1	129.4	6.8
Populus:Picea	80 years	1,100	0.1	0.9	240.2	143.1	26.2	139.6	124.5	138.9	-28.4
Experimental grassland mixtures (density and aboveground biomass yield, stems and grams per pot) (Mahaut et al., 2020)											
Bromus: Dactylis	13 weeks	6	0.5	0.5	5.5	9.8	2.6	8.6	5.3	11.5	3.8
Bromus: Lotus	13 weeks	6	0.5	0.5	5.5	6.3	3.8	3.2	5.3	4.7	-1.2
Bromus:Plantago	13 weeks	6	0.5	0.5	5.5	8.6	1.9	10.9	5.3	16.1	9.0
Bromus:Sanguisorba	13 weeks	6	0.5	0.5	5.5	4.4	2.9	2.2	5.3	3.6	0.3
Bromus:Trifolium	13 weeks	6	0.5	0.5	5.5	13.3	3.8	9.3	5.3	14.2	4.8
Dactylis:Lotus	13 weeks	6	0.5	0.5	9.8	6.3	11.8	2.0	11.5	4.7	3.4
Dactylis:Plantago	13 weeks	6	0.5	0.5	9.8	8.6	4.4	9.5	11.5	16.1	2.3
Dactylis:Sanguisorba	13 weeks	6	0.5	0.5	9.8	4.4	6.0	1.9	11.5	3.6	4.4
Dactylis:Trifolium	13 weeks	6	0.5	0.5	9.8	13.3	7.5	7.7	11.5	14.2	2.6
Lotus:Plantago	13 weeks	6	0.5	0.5	6.3	8.6	1.0	16.2	4.7	16.1	8.6
Lotus:Sanguisorba	13 weeks	6	0.5	0.5	6.3	4.4	1.9	2.6	4.7	3.6	-0.6
Lotus:Trifolium	13 weeks	6	0.5	0.5	6.3	13.3	2.4	9.2	4.7	14.2	4.4
Plantago:Sanguisorba	13 weeks	6	0.5	0.5	8.6	4.4	11.6	1.1	16.1	3.6	9.6
Plantago:Trifolium	13 weeks	6	0.5	0.5	8.6	13.3	10.2	6.9	16.1	14.2	3.3
Sanguisorba:Trifolium	13 weeks	6	0.5	0.5	4.4	13.3	1.9	13.4	3.6	14.2	5.3

Species codes: Populus - *Populus tremuloides,* Picea - *Picea glauca*, Bromus - *Bromus erectus,* Dactylis - *Dactylis glomerata*, Lotus - *Lotus corniculatus*, Plantago - *Plantago lanceolata*, Sanguisorba - *Sanguisorba minor*, and Trifolium - *Trifolium repens*.

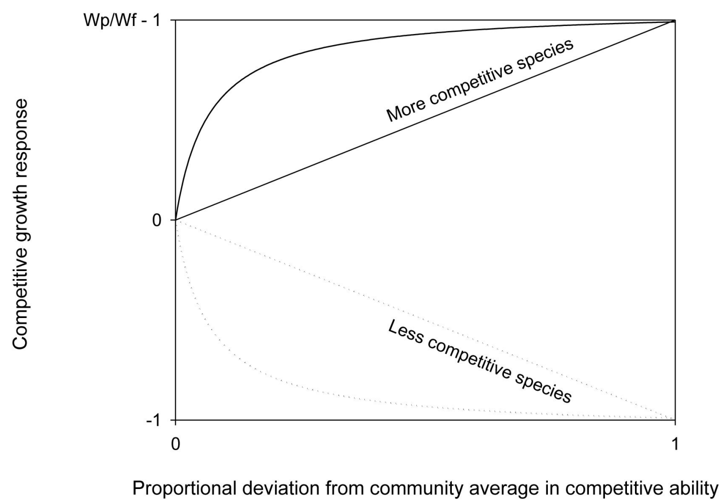

Figure 1.

Under the competitive hypothesis, more competitive species gain size (biomass or volume) and less competitive species lose size in mixture relative to their full density monocultures. The magnitude of competitive growth responses (i.e., proportional changes in individual size) increases with relative competitive ability (proportional deviations of species competitive ability from community average) from a minimum of 0 (at community average) to a maximum of $W_p/W_f - 1$ for more competitive species and -1 for less competitive species in competitive exclusion. The changes may not be linear, greater near community average and smaller with deviations of competitive ability (Gaudet & Keddy, 1988) or recourse availability (e.g., light, see Brüllhardt *et al*. 2020) away from community averages. W_f represents for individual size in full density monocultures and W_p represents for individual size in partial density monocultures.

Competitive expectation

Under the null hypothesis, species in mixture are assumed to be competitively equivalent (i.e., absence of interspecific interactions) and species growth (or any other function) stays the same in mixture as in monocultures (Loreau and Hector, 2001). Deviations of observed yields (biomass or volume) from total null expectation therefore produce net biodiversity effects or changes in ecosystem productivity (Loreau and Hector, 2001; Barry *et al*., 2019).

$$\begin{aligned}\sum Y_{Oi} - \sum Y_{NEi} &= \sum RY_{Oi}M_{if} - \sum RY_{Ei}M_{if} \\ &= \sum \Delta RY_i M_{if} \qquad (1)\end{aligned}$$

Thus, the net biodiversity effect is the difference between total observed yields $\sum Y_{Oi}$ and total null expectation $\sum Y_{NEi}$, calculated from species composition (or relative yield) in mixture RY_{Ei} and full density monoculture yield M_{if}. Mathematically, the net biodiversity effect is an arithmetic mean of full density monoculture yields weighted by changes in species relative yield ΔRY_I, and results from all species interactions including positive (positive biodiversity effects), competitive (positive or negative biodiversity effects), and negative (negative biodiversity effects) interactions. Net biodiversity effects can be mathematically decomposed into an arithmetic mean (a product of mean monoculture yield by mean relative yield change) representing community average response (CE) and a covariance (between monoculture yield and relative yield change) representing interspecies differences (SE) (Loreau and Hector, 2001).

To separate effects of competitive interactions from those of other species interactions, we would need the hypothesis that constituent species share an identical niche but differ in growth and competitive ability (i.e., absence of positive/negative interactions). Competitive growth responses, i.e., proportional changes in individual size (biomass or volume) expected from full density monocultures to mixtures under this competitive hypothesis, can be determined from species differences in growth and competitive ability (***Figure 1***). The magnitudes of competitive growth responses depend on: (1) species resource needs in full density monocultures; and (2) changes in resource availability (resources per capita) from full density monocultures to full density mixtures that are related to species composition and relative ability to compete for resources in mixture.

To estimate competitive growth response, we first determine the maximum competitive growth response MG_i that species can reach in a scenario of competitive exclusion (Murray, 1986) where a more competitive species completely dominates and grows like a partial density monoculture, while a less competitive species is competitively eliminated.

$$\begin{aligned}MG_i &= \frac{w_{ip}}{w_{if}} - 1\text{, for more competitive species} \\ &= -1\text{, for less competitive species} \qquad (2)\end{aligned}$$

Here, w_{ip}stands for individual size (biomass or volume) in partial density (species density in mixture) monoculture and w_{if} is for individual size in full density monoculture; both parameters are calculated from species monoculture yields and initial densities in partial/full density monocultures. A more competitive species gains $\frac{1}{RY_{Ei}} - 1$ in resource availability and $\frac{w_{ip}}{w_{if}} - 1$ in individual size, while a less competitive species loses 100% resource availability and size in mixture relative to full density monoculture (***Figure 1***). This simplified competitive dominance applies to mixtures where constituent species differ considerably in competitive ability, especially when competition for light is intense (Grime, 1973; Lieffers et al., 1993). In most cases, however, competitive exclusion would not occur in biodiversity experiments that are controlled in density, composition, duration, and scale. More competitive species are negatively affected by interspecific competition relative to their partial density monocultures (Holland and DeAngelis, 2009), while less competitive species are competitively suppressed relative to their null expectations, but not eliminated (Aarssen, 1989; Connolly et al., 2001; Montazeaud et al., 2018; Mahaut et al., 2020).

Next, we use relative competitive ability RC_i to represent species ability to compete for resources and achieve their maximum competitive growth response in mixture (Aarssen, 1989). RC_i is defined by the deviation of species competitive ability H_i from community average $\overline{H}_{i\notin}$ (excluding species i) relative to community maximum H_m. Competitive ability is species ability to exploit resources and can be assessed by growth attributes such as full density monoculture yields in biomass or volume (Gaudet and Keddy, 1988; Aarssen, 1989; Grace, 1990; Aschehoug et al., 2016; Montazeaud et al., 2018).

$$\mathrm{RC_i} = \frac{|H_i - \overline{H}_{i\notin}|}{H_m} \tag{3}$$

Greater competitive advantages would lead to increasingly positive competitive growth responses and larger competitive disadvantages would lead to increasingly negative competitive growth responses in mixture relative to full density monocultures.

$$\mathrm{Y_{CEi}} = \mathrm{Y_{NEi}}[1 + \mathrm{MG_i RC_i}] \tag{4}$$

Species competitive expectation Y_{CEi} is the yield based on null expectation Y_{NEi} and competitive growth response MG_iRC_iderived from species differences in growth and competitive ability (***Figure 1***). The term MG_iRC_iwould be positive for more competitive species $MG_i > 0$, negative for less competitive species $MG_i < 0$, and zero when species are at community average or competitively equivalent $RC_i = 0$. At individual species level, the deviation of observed yield Y_{Oi} from the competitive expectation Y_{CEi} can be positive (representing relaxation of interspecific competition from competitive expectation and therefore dominance of positive interactions), neutral (representing similar intra- and interspecific interactions or offset of positive and negative interactions), or negative (representing dominance of negative interactions). The differences between competitive and null expectations are competitive yield changes resulting from competitive interactions. Isolating facilitative effect from those of other positive interactions (i.e., resource partitioning) is also possible for individual species by comparing species' observed yield in mixture to their partial density monoculture yield, adapting the principle of additive design for biodiversity–ecosystem productivity experiments (Sackville Hamilton, 1994). At the community level, the difference in total yield between competitive expectations $\sum Y_{CEi}$ and null expectations $\sum Y_{NEi}$ of all species can be attributed to the competitive effect, whereas the difference between observed yields $\sum Y_{Oi}$and competitive expectations $\sum Y_{CEi}$ can be attributed to the dominance of positive effects when the difference >0 or to the dominance of negative effects when the difference <0 (***Figure 1***).

Case studies

Simulated aspen and spruce mixtures

Mixed forests composed of trembling aspen (*Populus tremuloides* Michx.) and white spruce (*Picea glauca* (Moench) Voss) are widely distributed in North America (Man and Lieffers, 1999). The two species, occurring in pure and mixed stands of varying compositions in natural conditions, generally have different canopy structure, rooting depth, leaf phenology, and requirements for light, moisture, and nutrients so are considered different in niche (Man and Lieffers, 1999).

We used a growth and yield simulation system GYPSY, developed for mixed species forests in western Canada (Huang et al., 2009), to generate data required for determining competitive expectation, i.e., mixtures and monocultures at partial and full densities. We were interested in knowing how effects of species interactions may vary with stand age and composition. We chose four ages: 20, 40, 60, and 80 years, to represent young, middle-aged, pre-mature, and mature stands, and five relative yields from nearly pure aspen (90% aspen), aspen-dominated (70% aspen), equal mixed (50% aspen), spruce-dominated (30% aspen), to nearly pure spruce (10% aspen) mixtures at average densities on fully stocked medium productivity sites (Peterson and Peterson, 1992) (***Table 1***). As GYPSY is developed from extensive field observations (Huang et al., 2009), the simulated data represent average yields in natural monocultures and mixtures of various composition and ages that may result from direct or indirect interactions between the two species (Man and Lieffers, 1999; Barry et al., 2019). Species competitive ability was approximated with stand volume in full density monocultures at site index 17 m and 14 m (height), respectively, for dominant and codominant aspen and spruce at 50 years of total age on medium productivity sites (Alberta Forest Service, 1985; Huang et al., 1994).

The partitioning of net biodiversity effects with competitive partitioning model indicated that the relative contribution of competitive interactions to net biodiversity effects decreased with age (***Figure 2a***), whereas community positive effects occurred generally in pre-mature and mature stands and more so in equal aspen and spruce mixtures (***Figure 2b***). Evidence for a facilitative effect (i.e., species observed yield in mixture > partial density monoculture yield) was only found on spruce in nearly pure mature spruce mixture (***Table 1***). Community negative effects occurred in young and middled-age mixtures where aspen did poorer than expected under the competitive expectation (***Table S1***). Averaged across all ages and compositions, the net biodiversity effect was 14.3 m^3 ha^{-1} or 14% of the average full density monoculture yield. On a relative basis, 57% of the value (8.2 m^3 ha^{-1}) was attributable to positive effects, 65% (9.3 m^3 ha^{-1}) to competitive effects and -23% (-3.2 m^3 ha^{-1}) to negative effects, compared to 63% and 37% for CE and SE with additive partitioning (***Figure 2c; Figure 2d***).

Experimental grassland mixtures

Mahaut et al. (2020) conducted an experimental study in pots with six grassland species in full and half-density monocultures, as well as in every combination of two, three and six species in mixture. The studied species vary in productivity and functional traits. We used aboveground biomass data from full and half-density monocultures, as well as all combinations of two species in full density mixtures (***Table 1***). Species competitive ability was approximated with species biomass in full density monocultures.

Competitive partitioning suggested a weak facilitative effect in two mixtures that were associated with *Lotus corniculatus*, one of the two legume species studied (***Table 1; Table S2***). Community negative effects occurred in two mixtures with *Sanguisorba minor* and between the two legumes, due to negative effects of *Sanguisorba minor* on other species and *Lotus corniculatus* on *Trifolium repens* (***Table S2***). Across all 15 two-species mixtures, the average of net biodiversity effects on aboveground biomass was 3.9 g/pot or 49% of the full density monoculture yield averaged across all species. On a relative basis, 60% (2.3 g) of the value was attributable to positive effects, 45% (1.7 g) to competitive effects, and -5% (-0.2 g) to negative effects, compared to 84% and 16% for CE and SE with additive partitioning (***Table 2***).

Comparison of additive and competitive partitioning

The competitive partitioning model provides a conceptual framework that helps establish the linkage between changes in ecosystem productivity and effects of species interactions. Under this framework, in both simulated tree and experimental grassland mixtures, net biodiversity effects were generally positive, resulting from both positive and competitive interactions. The positive competitive effects result from greater competitive yield responses of more competitive species

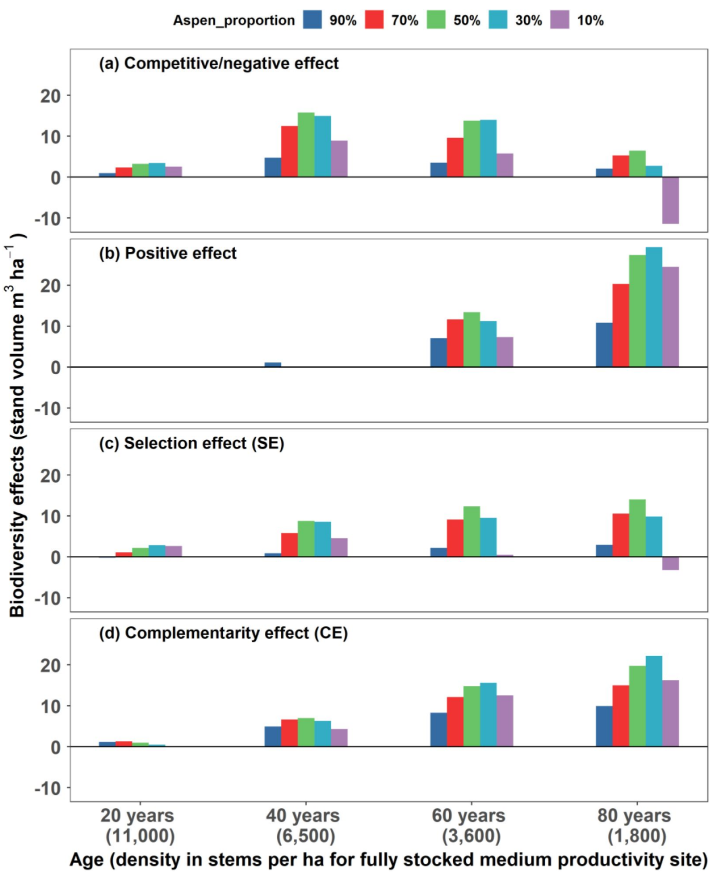

Figure 2.

Partitioning net biodiversity effects (changes in stand volume in mixtures relative to monocultures) with competitive (a and b) and additive (c and d) partitioning models for 20- to 80-year-old mixed trembling aspen and white spruce with varying species compositions from nearly pure aspen (90% aspen) to nearly pure spruce (90% spruce) on average medium productivity sites in western Canada. The growth and yield data (***Table 1***) are generated with GYPSY (Huang et al., 2009) and calculations of biodiversity components with this figure are detailed in ***Table S1***.

Species mixture	Competitive partitioning		Additive partitioning	
0.50Species1: 0.50Species2	Positive effect	Competitive/ negative effect	Complementarity effect (CE)	Selection effect (SE)
Bromus: Dactylis	1.8	1.7	2.7	0.9
Bromus: Lotus	1.2	-0.1	1.1	-0.1
Bromus:Plantago	2.5	3.2	4.3	1.4
Bromus:Sanguisorba	0.1	0.1	0.1	0.0
Bromus:Trifolium	0.9	2.8	3.6	0.0
Dactylis:Lotus	4.5	1.2	4.2	1.6
Dactylis:Plantago	4.3	0.3	5.1	-0.4
Dactylis:Sanguisorba	0.0	0.8	0.3	0.5
Dactylis:Trifolium	3.0	0.7	4.0	-0.3
Lotus:Plantago	7.6	2.3	7.9	1.9
Lotus:Sanguisorba	0.0	-0.9	-0.6	-0.3
Lotus:Trifolium	0.0	1.8	0.8	1.1
Plantago:Sanguisorba	1.5	4.7	3.9	2.3
Plantago:Trifolium	5.0	1.2	7.8	-1.6
Sanguisorba:Trifolium	2.9	3.6	3.9	2.6
Average	2.3	Competitive: 1.7 Negative: -0.2	3.3	0.6

Species codes: See ***Table 1***.

Table 2.

Partitioning net biodiversity effects (changes in aboveground biomass, g) with competitive and additive partitioning models using experimental data (**Table 1**) from a grassland diversity–productivity study (Mahaut et al., 2020). Calculations are detailed in ***Table S2***.

(Huston, 1997; Tilman et al., 1997; Wardle, 1999; Montazeaud et al., 2018), i.e., the yield gain of more competitive species from increases in resource availability exceeds the yield loss of less competitive species from decrease in resource availability, due to different competitive growth responses and full density monoculture yields (***Table S1; Table S2***). The greater competitive yield response of more competitive trembling aspen in young and middle-aged mixtures resulted in net biodiversity effects that were predominantly from competitive interactions (***Figure 2a; Figure 2b; Table S1***). A facilitative effect was rarely detected, only on spruce in nearly pure spruce mature mixture, possibly due to improved nutrient availability by aspen (Peterson and Peterson, 1992; Man and Lieffers, 1999) and in mixtures with legume *Lotus corniculatus* in grassland mixtures, likely due to improved nitrogen availability (Mahaut et al., 2020). The strong negative effect of mixing on aspen (i.e., observed yields < competitive expectations, see ***Table S1***) in young and middled-age mixtures suggests that trembling aspen originating naturally in high density does not benefit from mixing with white spruce. This is in contrast with white spruce where the deviations of observed yields from competitive expectations were generally positive, likely due to its shade tolerance and positive effects of aspen on nutrient availability (Peterson and Peterson, 1992; Man and Lieffers, 1999).

The net biodiversity effects reveal overall community yield changes from what would be expected from full density monoculture yields (Loreau and Hector, 2001; Loreau and Hector, 2019). However, the underlying mechanisms of species interactions for detected changes are uncertain, due to effects of competitive interactions that can be positive or negative. In both simulated mixed forests and experimental grassland mixtures, the partial density monoculture yield of the more competitive species exceeded the total expected yield of both species in nearly all mixtures (see ***Table S1*** and ***Table S2***), resulting in positive biodiversity effects when positive species interactions are not involved. Such positive biodiversity effects may help understand changes in ecosystem productivity detected through the null expectation (Carroll et al., 2011; Godoy et al., 2020) but do not support the general expectation that positive changes in ecosystem productivity are associated with positive species interactions (Loreau, 2000; Drake, 2003; Hooper et al., 2005; Barry et al., 2019) or crop mixing in agriculture and forestry where interests are positive effects (Man & Lieffers, 1999; Forrester & Pretzsch, 2015; Montazeaud et al., 2018). Thus, partitioning net biodiversity effects by effects of species interactions (i.e., positive, competitive, negative effects) allows for estimating relative importance of mechanisms of species interactions responsible for changes in ecosystem productivity. This enables ecologists to determine which communities are truly benefitting from diversity in the form of positive interactions and which result from interspecies differences in growth and competitive ability. In the aspen and spruce mixtures examined, community positive effects occurred largely in pre-mature and mature mixtures (>=60 years), even though net biodiversity effects and CE were positive at nearly all stages of stand development (***Figure 2***). In grassland mixtures, two mixtures (i.e., *Dactylis glomerata - Sanguisorba minor* and *Lotus corniculatus - Trifolium repens*) had positive net biodiversity effects that resulted entirely from competitive interactions (***Table 2; Table S2***). These positive net biodiversity effects may indicate evidence for positive changes in ecosystem productivity when these changes were actually driven by differential outcomes of interspecific competition.

We demonstrate with simulated and experimental species mixtures that competitive interaction is a major source of the net biodiversity effects and affects the additive components of both CE and SE. In the scenario of partial density monocultures emulating a simplified scenario of competitive exclusion where positive net biodiversity effects resulted entirely from competitive interactions, CE was generally negative in tree mixtures where relative yield changes were smaller with more competitive species (negative mean relative yield change) but positive in grassland mixtures where relative yield changes were greater with more competitive species (positive mean relative yield change) (***Table S1; Table S2***). In aspen-spruce mixtures, SE was either larger (age 80) or smaller (age 20, 40, and 60) than competitive effect. In the grassland communities, SE was similar to competitive effect in 3 of the 15 mixtures (*Bromus erectus - Lotus corniculatus*, *Dactylis* *glomerata* - *Lotus corniculatus*, and *Lotus corniculatus* - *Sanguisorba minor*) but substantially smaller in the remaining mixtures (***Table 2***). Even in the extreme cases where SE was nearly zero (*Bromus erectus* - *Sanguisorba minor* and *Bromus erectus* - *Trifolium repens*), competitive effect was still relatively strong, suggesting that a common assumption to attribute positive net biodiversity effects to positive species interactions when SE is close to zero (Loreau and Hector, 2001; Feng et al., 2022) is not justified. Competitive interactions amplify interspecies differences and therefore SE, while species interference on more competitive species (*Bromus erectus* - *Sanguisorba minor*) and greater positive species interactions on less competitive species (*Bromus erectus* - *Trifolium repens*) (***Table S2***) reduce SE. Our findings, along with some recent suggestions (Carroll et al., 2011; Turnbull et al., 2013; Godoy et al., 2020; Bourrat et al., 2023), do not support some common exercises that attribute positive/negative CEs to positive/negative interactions (Loreau and Hector, 2001; Petchey, 2003; Fargione et al., 2007; Barry et al., 2019; Hagan et al., 2021; Feng et al., 2022) and positive SEs to competitive interactions (Loreau, 2000; Loreau and Hector, 2001; Montazeaud et al., 2018).

Several assumptions are adopted with the development of competitive partitioning model. First, we assumed that more productive species are more competitive, which was mostly true with the species mixtures examined in terms of species yields in mixture relative to their full density monoculture yields (***Table 1***). There were exceptions with Dactylis:Plantago and Plantago:Trifolium in grassland mixtures where less productive species Plantago based on full density monoculture yields performed better in mixtures, which may suggest positive effects on Plantago and negative effects on the other species. However, species yields in partial versus full density monocultures indicate that these exceptions resulted from the stagnation of more productive Plantago in full (high) density monocultures, due to restricted rooting volumes and supplies of water and nutrients in pots (Chung et al., 2021). The release of growth stagnation in partial density monocultures and mixtures affects not only the mechanisms of changes in ecosystem productivity assessed with competitive and additive partitioning models but also the magnitudes of the changes in ecosystem productivity detected through the null expectation (Loreau and Hector, 2001). In this case, the use of partial density monoculture yields in assessment of species competitive ability are probably more appropriate, which would increase the values of competitive effects and reduce those of positive effects based on the competitive expectations. Partial density monocultures help detect competition-related yield changes, which would be otherwise unknown. Second, we used partial density monocultures to determine maximum competitive growth responses in a simplified competitive exclusion where more competitive species are not affected and grow like partial density monocultures, while less competitive species are competitively eliminated. We assumed a linear relationship between competitive growth responses and species relative competitive ability (***Figure 1***), which should be a good approximation (Gaudet and Keddy, 1988), but may not be the most appropriate, e.g., light availability and plant height (Brüllhardt et al., 2020). Linear models can overestimate or underestimate competitive expectations of all species and affect individual species assessments but would have limited impacts on community level assessments, due to different directions of competitive growth responses among species.

The competitive partitioning model incorporates effects of competitive interactions into the conventional null expectation and assists: (1) understanding the mechanisms of species interactions driving positive biodiversity-productivity relationships (Cardinale et al., 2012; Isbell et al., 2017) by relative contributions of positive interactions from more species of diverse niche (Tilman et al., 1997; Hooper et al., 2005) and competitive interactions from greater yield responses of more competitive and more productive species to changes in resource availability in mixture (Aarssen, 1997; Huston, 1997; Tilman et al., 1997; Pillai and Gouhier, 2019), (2) examining the mechanisms of key species in influencing ecosystem productivity by roles of competitive interactions due to their higher productivity and competitive advantages (Loreau and Hector, 2001; Mahaut et al., 2020), (3) meaningful comparisons of changes in ecosystem productivity (i.e., relative yield totals) across different ecosystems based on competitive expectations, and (4) redefining changes in ecosystem productivity such that positive changes result from positive interactions and negative changes from negative interactions (Loreau, 2000; Drake, 2003; Hooper et al., 2005). Under this competitive partitioning framework, null expectations are either raised for more competitive species with increased resource availability or lowered for less competitive species with decreased resource availability in mixture relative to full density monocultures. The magnitudes of competitive growth responses are derived from species relative competitive ability in mixture and their maximum competitive growth responses determined from full and partial density monocultures under a simplified scenario of competitive exclusion. Current biodiversity experiments do not have partial density monocultures and therefore do not provide estimates of competitive growth responses that would occur in mixture due to interspecies differences in growth and competitive ability. However, density-size/yield relationship is one of the most extensively studied areas in ecology (Watkinson, 1980; Weiner and Freckleton, 2010). Previous research can help determine maximum competitive growth responses through density-size/yield relationships (Huang et al., 2009). This means that the competitive partitioning model can be used to investigate changes in ecosystem productivity by effects of species interactions in current biodiversity experiments established with replacement design (Loreau and Hector, 2001; Fargione et al., 2007).

Conclusions

Competitive interactions are the major source of biodiversity effects and affect the additive components of both CE and SE. The interpretations of CEs and SEs with specific effects of species interactions lack mathematical or ecological justifications. For example, attributing positive biodiversity effects solely to effects of positive interactions, as commonly seen in the literature (Loreau and Hector, 2001; Hooper et al., 2005; Fargione et al., 2007; Barry et al., 2019; Feng et al., 2022), assumes that competitive interactions are productivity-neutral (SE close to zero) (Loreau and Hector, 2001; Feng et al., 2022). This assumption is true if ecosystem productivity is defined by total resources pool (Tilman et al., 2014) but generally not true if mixture productivity is defined by species monoculture yields as with the additive partitioning (Loreau and Hector, 2001), due to the predominance of competitive interactions in interspecific relationships (Goldberg and Barton, 1992) and positive effects of competitive interactions on ecosystem productivity based on the null expectation.

The competitive partitioning model is based on ecological theories of interspecific relationships, competitive ability, and competitive exclusion principle and experimental designs of both replacement and additive series. This new partitioning approach enables meaningful assessments of species interactions at both species and community levels and provides detailed insights into the mechanisms of species interactions that drive changes in ecosystem productivity. We believe that our framework, which is admittedly perfectible, is one promising avenue to determine effects of species interactions responsible for changes in ecosystem productivity, an approach that is long sought after in biodiversity–ecosystem productivity research (Loreau and Hector, 2019; Mahaut et al., 2020).

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Author contributions

R.M. and H.Y. designed the research; J.T., S.H. and R.M. generated growth and yield data with the GYPSY; and J.T., C.A.N., E.B.S., S.H., R.M., H.Y., G.T.F., C.V., and J.Z. contributed to the interpretations of the results and to the writing of the paper.

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Editors

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Reviewer #1 (Public Review):

[Editors' note: this is an overall synthesis from the Reviewing Editor in consultation with the reviewers.]

The three reviews expand our critique of this manuscript in some depth and complementary directions. These can be synthesized in the following main points (we point out that there is quite a bit more that could be written about the flaws with this study; however, time constraints prevented us from further elaborating on the issues we see):

(1) It is unclear what the authors want to do. It seems their main point is that the large BEF literature and especially biodiversity experiments overstate the occurrence of positive biodiversity effects because some of these can result from competition. Because reduced interspecific relative to intraspecific competition in mixture is sufficient to produce positive effects in mixtures (if interspecific competition = 0 then RYT = S, where S is species richness in mixture -- this according to the reciprocal yield law = law of constant final yield), they have a problem accepting NE > 0 as true biodiversity effect (see additive partitioning method of Loreau & Hector 2001 cited in manuscript).

(2) The authors' next claim, without justification, that additive partitioning of NE is flawed and theoretically and biologically meaningless. They misinterpret the CE component as biological niche partitioning and the SE component as biological dominance. They do not seem to accept that the additive partitioning is a logically and mathematically sound derivation from basic principles that cannot be contested.

(3) The authors go on to introduce a method to calculate species-level overyielding (RY > 1/S in replacement series experiments) as a competitive growth response and multiply this with the species monoculture biomass relative to the maximum to obtain competitive expectation. This method is based on resource competition and the idea that resource uptake is fully converted into biomass (instead of e.g. investing it in allelopathic chemical production).

(4) It is unclear which experiments should be done, i.e. are partial-density monocultures planted or simply calculated from full-density monocultures? At what time are monocultures evaluated? The framework suggests that monocultures must have the full potential to develop, but in experiments, they are often performing very poorly, at least after some time. I assume in such cases the monocultures could not be used.

(5) There are many reasons why the ideal case of only resource competition playing a role is unrealistic. This excludes enemies but also differential conversion factors of resources into biomass and antagonistic or facilitative effects. Because there are so many potential reasons for deviations from the null model of only resource competition, a deviation from the null model does not allow conclusions about underlying mechanisms.

Furthermore, this is not a systematically developed partitioning, but some rather empirical ad hoc formulation of a first term that is thought to approximate competitive effects as understood by the authors (but again, there already are problems here). The second residual term is not investigated. For a proper partitioning approach, one would have to decompose overyielding into two (or more) terms and demonstrate (algebraically) that under some reasonable definitions of competitive and non-competitive interactions, these end up driving the respective terms.

(6) Using a simplistic simulation to test the method is insufficient. For example, I do not see how the simulation includes a mechanism that could create CE in additive partitioning if all species would have the same monoculture yield. Similarly, they do not include mechanisms of enemies or antagonistic interactions (e.g. allelopathy).

(7) The authors do not cite relevant literature regarding density x biodiversity experiments, competition experiments, replacement-series experiments, density-yield experiments, additive partitioning, facilitation, and so on.

Overall, this manuscript does not lead further from what we have already elaborated in the broad field of BEF and competition studies and rather blurs our understanding of the topic.

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Reviewer #2 (Public Review):

This manuscript is motivated by the question of what mechanisms cause overyielding in mixed-species communities relative to the corresponding monocultures. This is an important and timely question, given that the ultimate biological reasons for such biodiversity effects are not fully understood.

As a starting point, the authors discuss the so-called "additive partitioning" (AP) method proposed by Loreau & Hector in 2001. The AP is the result of a mathematical rearrangement of the definition of overyielding, written in terms of relative yields (RY) of species in mixtures relative to monocultures. One term, the so-called complementarity effect (CE), is proportional to the average RY deviations from the null expectations that plants of both species "do the same" in monocultures and mixtures. The other term, the selection effect (SE), captures how these RY deviations are related to monoculture productivity. Overall, CE measures whether relative biomass gains differ from zero when averaged across all community members, and SE, whether the "relative advantage" species have in the mixture, is related to their productivity. In extreme cases, when all species benefit, CE becomes positive. When large species have large relative productivity increases, SE becomes positive. This is intuitively compatible with the idea that niche complementarity mitigates competition (CE>0), or that competitively superior species dominate mixtures and thereby driver overyielding (SE>0).

However, it is very important to understand that CE and SE capture the "statistical structure" of RY that underlies overyielding. Specifically, CE and SE are not the ultimate biological mechanisms that drive overyielding, and never were meant to be. CE also does not describe niche complementarity. Interpreting CE and SE as directly quantifying niche complementarity or resource competition, is simply wrong, although it sometimes is done. The criticism of the AP method thus in large part seems unwarranted. The alternative methods the authors discuss (lines 108-123) are based on very similar principles.

The authors now set out to develop a method that aims at linking response patterns to "more true" biological mechanisms.

Assuming that "competitive dominance" is key to understanding mixture productivity, because "competitive interactions are the predominant type of interspecific relationships in plants", the authors introduce "partial density" monocultures, i.e. monocultures that have the same planting density for a species as in a mixture. The idea is that using these partial density monocultures as a reference would allow for isolating the effect of competition by the surrounding "species matrix".

The authors argue that "To separate effects of competitive interactions from those of other species interactions, we would need the hypothesis that constituent species share an identical niche but differ in growth and competitive ability (i.e., absence of positive/negative interactions)." - I think the term interaction is not correctly used here, because clearly competition is an interaction, but the point made here is that this would be a zero-sum game.

The authors use the ratio of productivity of partial density and full-density monocultures, divided by planting density, as a measure of "competitive growth response" (abbreviated as MG). This is the extra growth a plant individual produces when intraspecific competition is reduced.

Here, I see two issues: first, this rests on the assumption that there is only "one mode" of competition if two species use the same resources, which may not be true, because intraspecific and interspecific competition may differ. Of course, one can argue that then somehow "niches" are different, but such a niche definition would be very broad and go beyond the "resource set" perspective the authors adopt. Second, this value will heavily depend on timing and the relationship between maximum initial growth rates and competitive abilities at high stand densities.

The authors then progress to define relative competitive ability (RC), and this time simply uses monoculture biomass as a measure of competitive ability. To express this biomass in a standardized way, they express it as different from the mean of the other species and then divide by the maximum monoculture biomass of all species.

I have two concerns here: first, if competitive ability is the capability of a species to preempt resources from a pool also accessed by another species, as the authors argued before, then this seems wrong because one would expect that a species can simply be more productive because it has a broader niche space that it exploits. This contradicts the very narrow perspective on competitive ability the authors have adopted. This also is difficult to reconcile with the idea that specialist species with a narrow niche would outcompete generalist species with a broad niche. Second, I am concerned by the mathematical form. Standardizing by the maximum makes the scaling dependent on a single value.

As a final step, the authors calculate a "competitive expectation" for a species' biomass in the mixture, by scaling deviations from the expected yield by the product MG × RC. This would mean a species does better in a mixture when (1) it benefits most from a conspecific density reduction, and (2) has a relatively high biomass.

Put simply, the assumption would be that if a species is productive in monoculture (high RC), it effectively does not "see" the competitors and then grows like it would be the sole species in the community, i.e. like in the partial density monoculture.

Overall, I am not very convinced by the proposed method.

(1) The proposed method seems not very systematic but rather "ad hoc". It also is much less a partitioning method than the AP method because the other term is simply the difference. It would be good if the authors investigated the mathematical form of this remainder and explored its properties.. when does complementarity occur? Would it capture complementarity and facilitation?

(2) The justification for the calculation of MG and RC does not seem to follow the very strict assumptions of what competition (in the absence of complementarity) is. See my specific comments above.

(3) Overall, the manuscript is hard to read. This is in part a problem of terminology and presentation, and it would be good to use more systematic terms for "response patterns" and "biological mechanisms".

Examples:
- on line 30, the authors write that CE is used to measure "positive" interactions and SE to measure "competitive interactions", and later name "positive" and "negative" interactions "mechanisms of species interactions". Here the authors first use "positive interaction" as any

type of effect that results in a community-level biomass gain, but then they use "interaction" with reference to specific biological mechanisms (e.g. one species might attract a parasite that infests another species, which in turn may cause further changes that modify the growth of the first and other species).

- on line 70, the authors state that "positive interaction" increases productivity relative to the null expectation, but it is clear that an interaction can have "negative" consequences for one interaction partner and "positive" ones for the other. Therefore, "positive" and "negative" interactions, when defined in this way, cannot be directly linked to "resource partitioning" and "facilitation", and "species interference" as the authors do. Also, these categories of mechanisms are still simple. For example, how do biotic interactions with enemies classify, see above?

- line 145: "Under the null hypothesis, species in the mixture are assumed to be competitively equivalent (i.e., absence of interspecific interactions)". This is wrong. The assumption is that there are interspecific interactions, but that these are the same as the intraspecific ones. Weirdly, what follows is a description of the AP method, which does not belong here. This paragraph would better be moved to the introduction where the AP method is mentioned. Or omitted, since it is basically a repetition of the original Loreau & Hector paper.

Other points:

- line 66: community productivity, not ecosystem productivity.
- line 68: community average responses are with respect to relative yields - this is important!
- line 64: what are "species effects of species interactions" ?
- line 90: here "competitive" and "productive" are mixed up, and it is important to state that "suffers more" refers to relative changes, not yield changes.
- line 92: "positive effect of competitive dominance": I don't understand what is meant here.

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Reviewer #3 (Public Review):

Summary:

This manuscript by Tao et al. reports on an effort to better specify the underlying interactions driving the effects of biodiversity on productivity in biodiversity experiments. The authors are especially concerned with the potential for competitive interactions to drive positive biodiversity-ecosystem functioning relationships by driving down the biomass of subdominant species. The authors suggest a new partitioning schema that utilizes a suite of partial density treatments to capture so-called competitive ability. While I agree with the authors that understanding the underlying drivers of biodiversity-ecosystem functioning relationships is valuable - I am unsure of the added value of this specific approach for several reasons.

Strengths:

I can find a lot of value in endeavouring to improve our understanding of how biodiversity-ecosystem functioning relationships arise. I agree with the authors that competition is not well integrated into the complementarity and selection effect and interrogating this is important.

Weaknesses:

(1) The authors start the introduction very narrowly and do not make clear why it is so important to understand the underlying mechanisms driving biodiversity-ecosystem

functioning relationships until the end of the discussion.

(2) The authors criticize the existing framework for only incorporating positive interactions but this is an oversimplification of the existing framework in several ways:
a. The existing partitioning scheme incorporates resource partitioning which is an effect of competition.
b. The authors neglect the potential that negative feedback from species-specific pests and pathogens can also drive positive BEF and complementarity effects but is not a positive interaction, necessarily. This is discussed in Schnitzer et al. 2011, Maron et al. 2011, Hendriks et al. 2013, Barry et al. 2019, etc.
c. Hector and Loreau (and many of the other citations listed) do not limit competition to SE because resource partitioning is a byproduct of competition.

(3) It is unclear how this new measure relates to the selection effect, in particular. I would suggest that the authors add a conceptual figure that shows some scenarios in which this metric would give a different answer than the traditional additive partition. The example that the authors use where a dominant species increases in biomass and the amount that it increases in biomass is greater than the amount of loss from it outcompeting a subdominant species is a general example often used for a selection effect when exactly would you see a difference between the two? :
a. Just a note - I do think you should see a difference between the two if the species suffers from strong intraspecific competition and has therefore low monoculture biomass but this would tend to also be a very low-density monoculture in practice so there would potentially be little difference between a low density and high-density monoculture because the individuals in a high-density monoculture would die anyway. So I am not sure that in practice you would really see this difference even if partial density plots were incorporated.

(4) One of the tricky things about these endeavors is that they often pull on theory from two different subfields and use similar terminology to refer to different things. For example - in competition theory, facilitation often refers to a positive relative interaction index (this seems to be how the authors are interpreting this) while in the BEF world facilitation often refers to a set of concrete physical mechanisms like microclimate amelioration. The truth is that both of these subfields use net effects. The relative interaction index is also a net outcome as is the complementarity effect even if it is only a piece of the net biodiversity effect. Trying to combine these two subfields to come up with a new partitioning mechanism requires interrogating the underlying assumptions of both subfields which I do not see in this paper.

(5) The partial density treatment does not isolate competition in the way that the authors indicate. All of the interactions that the authors discuss are density-dependent including the mechanism that is not discussed (negative feedback from species-specific pests and pathogens). These partial density treatment effects therefore cannot simply be equated to competition as the authors indicate.:
a. Additionally - the authors use mixture biomass as a stand-in for competitive ability in some cases but mixture biomass could also be determined by the degree to which a plant is facilitated in the mixture (for example).

(6) I found the literature citation to be a bit loose. For example, the authors state that the additive partition is used to separate positive interactions from competition (lines 70-76) and cite many papers but several of these (e.g. Barry et al. 2019) explicitly do not say this.

(7) The natural take-home message from this study is that it would be valuable for biodiversity experiments to include partial density treatments but I have a hard time seeing this as a valuable addition to the field for two reasons:
a. In practice - adding in partial density treatments would not be feasible for the vast majority of experiments which are already often unfeasibly large to maintain.
b. The density effect would likely only be valuable during the establishment phase of the

experiment because species that are strongly limited by intraspecific competition will die in the full-density plots resulting in low-density monocultures. You can see this in many biodiversity experiments after the first years. Even though they are seeded (or rarely planted) at a certain density, the density after several years in many monocultures is quite low.

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Reviewer #4 (Public Review):

Summary:

This manuscript claims to provide a new null hypothesis for testing the effects of biodiversity on ecosystem functioning. It reports that the strength of biodiversity effects changes when this different null hypothesis is used. This main result is rather inevitable. That is, one expects a different answer when using a different approach. The question then becomes whether the manuscript's null hypothesis is both new and an improvement on the null hypothesis that has been in use in recent decades.

Strengths:

In general, I appreciate studies like this that question whether we have been doing it all wrong and I encourage consideration of new approaches.

Weaknesses:

Despite many sweeping critiques of previous studies and bold claims of novelty made throughout the manuscript, I was unable to find new insights. The manuscript fails to place the study in the context of the long history of literature on competition and biodiversity and ecosystem functioning. The Introduction claims the new approach will address deficiencies of previous approaches, but after reading further I see no evidence that it addresses the limitations of previous approaches noted in the Introduction. Furthermore, the manuscript does not reproducibly describe the methods used to produce the results (e.g., in Table 1) and relies on simulations, claiming experimental data are not available when many experiments have already tested these ideas and not found support for them. Finally, it is unclear to me whether rejecting the 'new' null hypothesis presented in the manuscript would be of interest to ecologists, agronomists, conservationists, or others. I will elaborate on each of these points below.

The critiques of biodiversity experiments and existing additive partitioning methods are overstated, as is the extent to which this new approach addresses its limitations. For example, the critique that current biodiversity experiments cannot reveal the effects of species interactions (e.g., lines 37-39) isn't generally true, but it could be true if stated more specifically. That is, this statement is incorrect as written because comparisons of mixtures, where there are interspecific and intraspecific interactions, with monocultures, where there are only intraspecific interactions, certainly provide information about the effects of species interactions (interspecific interactions). These biodiversity experiments and existing additive partitioning approaches have limits, of course, for identifying the specific types of interactions (e.g., whether mediated by exploitative resource competition, apparent competition, or other types of interactions). However, the approach proposed in this manuscript gets no closer to identifying these specific mechanisms of species interactions. It has no ability to distinguish between resource and apparent competition, for example. Thus, the motivation and framing of the manuscript do not match what it provides. I believe the entire Introduction would need to be rewritten to clarify what gap in knowledge this proposed approach is addressing and what would be gained by filling this knowledge gap.

I recommend that the Introduction instead clarify how this study builds on and goes beyond many decades of literature considering how competition and biodiversity effects depend on density. This large literature is insufficiently addressed in this manuscript. This fails to give credit to previous studies considering these ideas and makes it unclear how this manuscript goes beyond the many previous related studies. For example, see papers and books written by de Wit, Harper, Vandermeer, Connolly, Schmid, and many others. Also, note that many biodiversity experiments have crossed diversity treatments with a density treatment and found no significant effects of density or interactions between density and diversity (e.g., Finn et al. 2013 Journal of Applied Ecology). Thus, claiming that these considerations of density are novel, without giving credit to the enormous number of previous studies considering this, is insufficient.

Replacement series designs emerged as a consensus for biodiversity experiments because they directly test a relevant null hypothesis. This is not to say that there are no other interesting null hypotheses or study designs, but one must acknowledge that many designs and analyses of biodiversity experiments have already been considered. For example, Schmid et al. reviewed these designs and analyses two decades ago (2002, chapter 6 in Loreau et al. 2002 OUP book) and the overwhelming consensus in recent decades has been to use a replacement series and test the corresponding null hypothesis.

It is unclear to me whether rejecting the 'new' null hypothesis presented in the manuscript would be of interest to ecologists, agronomists, conservationists, or others. Most biodiversity experiments and additive partitions have tested and quantified diversity effects against the null hypothesis that there is no difference between intraspecific and interspecific interactions. If there was no less competition and no more facilitation in mixtures than in monocultures, then there would be no positive diversity effects. Rejecting this null hypothesis is relevant when considering coexistence in ecology, overyielding in agronomy, and the consequences of biodiversity loss in conservation (e.g., Vandermeer 1981 Bioscience, Loreau 2010 Princeton Monograph). This manuscript proposes a different null hypothesis and it is not yet clear to me how it would be relevant to any of these ongoing discussions of changes in biodiversity.

The claim that all previous methods 'are not capable of quantifying changes in ecosystem productivity by species interactions and species or community level' is incorrect. As noted above, all approaches that compare mixtures, where there are interspecific interactions, to monocultures, where there are no species interactions, do this to some extent. By overstating the limitations of previous approaches, the manuscript fails to clearly identify what unique contribution it is offering, and how this builds on and goes beyond previous work.

The manuscript relies on simulations because it claims that current experiments are unable to test this, given that they have replacement series designs (lines 128-131). There are, however, dozens of experiments where the replacement series was repeated at multiple densities, which would allow a direct test of these ideas. In fact, these ideas have already been tested in these experiments and density effects were found to be nonsignificant (e.g., Finn et al. 2013).

It seems that the authors are primarily interested in trees planted at a fixed density, with no opportunity for changes in density, and thus only changes in the size of individuals (e.g., Fig. 1). In natural and experimental systems, realized density differs from the initial planted density, and survivorship of seedlings can depend on both intraspecific and interspecific interactions. Thus, the constrained conditions under which these ideas are explored in this manuscript seem narrow and far from the more complex reality where density is not fixed.

Additional detailed comments:

It is unclear to me which 'effects' are referred to on line 36. For example, are these diversity effects or just effects of competition? What is the response variable?

The usefulness of the approach is overstated on line 52. All partitioning approaches, including the new one proposed here, give the net result of many types of species interactions and thus cannot 'disentangle underlying mechanisms of species interactions.'

The weaknesses of previous approaches are overstated throughout the manuscript, including in lines 60-61. All approaches provide some, but not all insights. Sweeping statements that previous approaches are not effective, without clarifying what they can and can't do, is unhelpful and incorrect. Also, these statements imply that the approach proposed here addresses the limitations of these previous approaches. I don't yet see how it does so.

The definitions given for the CE and SE on line 71 are incorrect. Competition affects both terms and CE can be negative or have nothing to do with positive interactions, as noted in many of the papers cited.

The proposed approach does not address the limitations noted on lines 73 and 74.

The definition of positive interactions in lines 77 and 78 seems inconsistent with much of the literature, which instead focuses on facilitation or mutualism, rather than competition when describing positive interactions.

Throughout the manuscript, competition is often used interchangeably with resource competition (e.g., line 82) and complementarity is often attributed to resource partitioning (e.g., line 77). This ignores apparent competition and partitioning enemy-free niche space, which has been found to contribute to biodiversity effects in many studies.

In what sense are competitive interactions positive for competitive species (lines 82-83)? By definition, competition is an interaction that has a negative effect. Do you mean that interspecific competition is less than intraspecific competition? I am having a very difficult time following the logic.

Results are asserted on lines 93-95, but I cannot find the methods that produced these results. I am unable to evaluate the work without a repeatable description of the methods.

The description of the null hypothesis in the common additive partitioning approach on lines 145-146 is incorrect. In the null case, it does not assume that there are no interspecific interactions, but rather that interspecific and intraspecific interactions are equivalent.

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